

15+ ADDITIONAL THREAT HUNTING USE CASES

These are professional SOC-level use cases you can use in labs or interviews.

Use Case 1 — Suspicious PowerShell Execution

Hypothesis: Attackers using encoded PowerShell commands.

Source: Sysmon, Windows Event 4104.

Use Case 2 — Brute Force Attack on Windows Logins

Multiple login failures followed by success.

Use Case 3 — Privilege Escalation via Token Manipulation

Sysmon Event ID 1 + 10 correlation.

Use Case 4 — RDP Lateral Movement

Abnormal RDP connections between internal machines.

Use Case 5 — Unusual Admin Account Logins

Logins from:

- Rare systems
- Odd hours
- New geographic locations

Use Case 6 — Suspicious Registry Persistence

Monitor:

- Run keys
- Services creation

Use Case 7 — DNS Tunneling Detection

Long domain queries & unusual domain entropy.

Use Case 8 — C2 Beacons Detection

Repeated connections to same external IP with fixed interval.

Use Case 9 — Rare Parent-Child Process

Examples:

- winword.exe → powershell.exe
- explorer.exe → cmd.exe
- excel.exe → rundll32.exe

Use Case 10 — Suspicious Scheduled Task Creation

Possible malware persistence mechanism.

Use Case 11 — Unauthorized VPN Login

Impossible travel detection.

Use Case 12 — SMB Lateral Spread (WannaCry-style)

Massive SMB connections to multiple hosts.

Use Case 13 — Data Exfiltration

Large outbound traffic to unknown external IP.

Use Case 14 — Web Shell Detection

Example:

- China Chopper patterns
- POST requests with suspicious headers

Use Case 15 — Suspicious File Execution from Temp Folder

Common in malware infections.

Use Case 16 — Suspicious Script Execution (JS/VBS) in Startup

Malware persistence through script engine.

Use Case 17 — Disabled Security Controls

Detection for:

- AV disabled
- EDR tampering
- Firewall disabled

Use Case 18 — Cloud Account Compromise

AWS / Azure abnormal activity:

- New access key
- Root login
- EC2 crypto mining behavior